

To: Dr. Marisol Vega (Primary Care)

From: Hematology/Oncology Service

Date: 16.04.2024

Patient: Joan Torres, MRN AML070

Diagnosis: Acute Myeloid Leukemia (AML), **ELN 2022 Favorable, inv(16)(p13.1q22)** consistent with **CBFB–MYH11** fusion

Initial therapy: Induction **7+3** (cytarabine + daunorubicin) **started 25.12.2022**, followed by high-dose cytarabine consolidation

Reason for letter

Transitioning Ms. Torres (*05.05.1979) from active chemotherapy to **surveillance** after sustained complete remission and molecular response. This document summarizes her diagnostic workup, treatment course, toxicity management, and survivorship plan.

Presentation and Diagnostic Workup

Ms. Torres presented on **18.12.2022** with a two-month history of progressive fatigue, gingival bleeding, and easy bruising. Initial labs: **WBC $17.9 \times 10^9/L$** (blasts 29%), **Hgb 7.5 g/dL**, **Plt $51 \times 10^9/L$** , **LDH 590 U/L**, uric acid 6.2 mg/dL, normal coagulation. Peripheral smear revealed myeloblasts and abnormal eosinophils.

Bone marrow biopsy (18.12.2022): 70–75% blasts with increased abnormal eosinophils; flow cytometry: **CD34⁺**, **CD117⁺**, **HLA-DR⁺**, **CD13⁺/CD33⁺**, partial **CD19** expression (seen in core-binding factor AML).

Cytogenetics: **inv(16)(p13.1q22)**; otherwise diploid.

Molecular panel: FLT3-ITD/TKD negative; NPM1 negative; KIT D816V negative.

Baseline echocardiogram: LVEF **63%**; QTc 431 ms.

Infectious screen: HIV/HBV/HCV negative; CMV IgG positive.

Treatment Summary

Induction “7+3” (25.12.2022):

- Cytarabine continuous infusion D1–7; daunorubicin D1–3.
- **Complications:** One episode of **febrile neutropenia** (cultures negative, resolved with piperacillin-tazobactam), grade-2 mucositis, and mild transaminitis (ALT peak 86 U/L). **No TLS.**
- **Day-28 marrow (23.01.2023): Complete remission (CR)** morphologic; blasts <2%. **CBFB–MYH11 RT-qPCR** demonstrated **>3-log reduction** from baseline.

Consolidation HiDAC (Feb–Jun 2023):

- **Dose:** Cytarabine **3 g/m² q12h D1,3,5** × 4 cycles.

- **Neuro-ophthalmic monitoring:** Daily cerebellar checks; ocular steroid drops each cycle—no neurotoxicity or conjunctivitis.
- **Infectious course:** One short hospitalization in April for rhinoviral FN; cultures sterile.
- **MRD kinetics:**
 - After cycle 2 (Mar 2023): **>4-log reduction** in CBFB–MYH11 transcript.
 - After cycle 4 (Jun 2023): **Undetectable** (assay sensitivity 10^{-5}).
- **Central line:** Right IJ tunnelled catheter placed 12/2022, removed 07/2023.

Post-consolidation (Jul 2023→present): No maintenance therapy given **favorable-risk** CBF-AML with sustained molecular clearance. Surveillance approach adopted.

Current Status (clinic 08.04.2024)

Symptoms/Function: Returned to her work as a physical therapist; runs 3–4 km three times weekly. Denies fever, infections, bleeding, or night sweats. ECOG 0.

Exam: Afebrile; BP 114/70; HR 72; clear lungs; no lymphadenopathy or organomegaly; skin clear of petechiae.

Laboratory data (08.04.2024):

- **CBC:** WBC $5.3 \times 10^9/L$ (ANC 2.7), Hgb **13.2 g/dL**, Plt $248 \times 10^9/L$; retic 1.2%.
- **CMP:** Na 139, K 4.2, Cr **0.84 mg/dL** (eGFR >90), AST 22, ALT 18, Tbili 0.5, albumin 4.2.
- **Inflammation:** CRP <1 mg/L; LDH 170 U/L.
- **MRD:** CBFB–MYH11 RT-qPCR **undetectable** (peripheral blood, 03.2024).
- **Cardiac:** Echo 03.2024 LVEF **61%**.

Comorbidities & Medications

- **Primary hypothyroidism (Hashimoto):** levothyroxine **50 µg qAM**; TSH 2.0 mIU/L.
- **Migraine without aura:** sumatriptan **25–50 mg PRN** (<2 doses/24 h).
- **Vitamin D insufficiency:** cholecalciferol 1000 IU daily.
- No anticoagulation; no diabetes or renal disease. NKDA.

Late-effects & Survivorship

- **Cardiotoxicity:** Anthracycline exposure tracked; next surveillance echo in **2026** or sooner if symptomatic.
- **Neuro-ocular:** No cytarabine toxicity; annual eye exams.
- **Bone health:** Baseline DEXA normal; continues weight-bearing exercise.
- **Fertility/gynecology:** Premenopausal; cycles regular; contraception via copper IUD.

- **Vaccination:** Influenza annually; COVID booster; pneumococcal series complete.
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Plan

1. **Surveillance only.** CBC and clinical review **q3 months** in 2024, then extend to **q4–6 months** if stable. **CBFB–MYH11 RT-qPCR (PB) q6 months** for two additional years.
2. **When to call:** $T \geq 38.0^{\circ}\text{C}$, mucosal bleeding/petechiae, persistent fatigue, night sweats, or new lymphadenopathy.
3. **Dental/procedures:** No special restrictions currently; if future thrombocytopenia develops, coordinate hemostasis.
4. **Lifestyle:** Continue exercise; avoid NSAIDs if any future thrombocytopenia; maintain vaccinations (non-live).

Summary: Ms. Torres remains in **ongoing complete remission** with **sustained molecular negativity** after standard induction and HiDAC consolidation for **CBF-AML with inv(16)**. No further chemotherapy is planned.

Warm regards,
[Name], MD — Consultant Hematologist